MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2019

Mathematical Methods

Module Leader: Dr P. Philipp and Professor K. Zhang

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) Find the critical point of

$$f(x, y) = xy - x,$$

and determine whether it is a local minimum, a local maximum, or a saddle point. (8)

(b) Express the definite integral

$$\int_0^9 (2x + 1)\,\mathrm{d}x$$

as a limit of Riemann sums and evaluate that limit. (8)

(c) Find the following integrals.

$$\text{(i)} \int x \cos x\,\mathrm{d}x, \quad \text{(ii)} \int_0^3 x^2\sqrt{9 - x^2}\,\mathrm{d}x. \tag{10}$$

(d) Solve the following ordinary differential equations.

(i) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{3}{x}y = 4 + x^2$ with $y(1) = \dfrac{7}{6},\ x > 0;$

(ii) $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + y = \mathrm{e}^{-x}$ with $y(0) = 1,\ y'(0) = 0.$ (12)

(e) Suppose that $f(x) = x^2 - 2x - 3$ and consider

$$A = \begin{bmatrix} -1 & 0 \\ 4 & 3 \end{bmatrix}.$$

Find the matrix $f(A)$. (6)

(f) Evaluate the following integral.

$$\int_{-1}^{1}\int_{-1}^{1} 3(x^2 + y^2)\,\mathrm{d}y\,\mathrm{d}x.$$

(6)

[**50**]

SECTION B

2. (a) Let

$$f(x) = \int_0^{\sin^2 x} \sin^{-1}\sqrt{t}\,\mathrm{d}t + \int_0^{\cos^2 x} \cos^{-1}\sqrt{t}\,\mathrm{d}t, \quad 0 < x < \frac{\pi}{2}.$$

Show that $f(x)$ is constant. (10)

(b) Evaluate the expression

$$I = \lim_{n\to\infty} \frac{1}{n}\left[(n+1)(n+2)(n+3)...(n+n)\right]^{1/n}$$

by associating it with a definite integral. (15)

[25]

3. (a) Find the values of a and b in

$$\xi = x + ay, \quad \eta = x + by,$$

such that switching from (x, y) to (ξ, η) transforms the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial^2 u}{\partial x \partial y} + 3\frac{\partial^2 u}{\partial x^2} = 0$$

to

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$$

(10)

(b) Find the general solution $x(t)$ of the third-order differential equation

$$t\frac{\mathrm{d}^3 x}{\mathrm{d}t^3} + 3\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} - t\frac{\mathrm{d}x}{\mathrm{d}t} - x = 0, \qquad t > 0.$$

[Hint: Consider $y(t) = t\,x(t)$.] (15)

[25]

4. (a) Find all the eigenvalues of the matrix

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

(13)

(b) Find the eigenvectors corresponding to those eigenvalues. (12)

[25]